

Around 85-90% of our components are manufactured/sourced locally, with only certain specialised electronic parts imported,” says Kirti. This combination of local manufacturing and advanced engineering makes Canebot a true ‘Made in India’ product.

Milind shares that they have secured seven patents for different mechanisms of Canebot, thereby creating a strong entry barrier. The startup is implementing a franchise model to build a brand around its technology. “This approach allows us to maintain control over the quality of sugarcane used, as the hygiene of the juice is heavily dependent on the raw material. Through this model, we provide machines to

micro-entrepreneurs and manage the supply chain by partnering with authorised suppliers to ensure high-quality inputs,” adds Kirti.

She also notes that their compact machines, which require only 1.11 square metre (12 square feet) of space, can be rapidly scaled and installed in airports and malls. This will help create thousands of micro-entrepreneurs while providing them with technical support, training, and raw materials.

The business model involves both one-time and recurring revenue streams. The one-time revenue is generated from the sale of franchises and the associated machines to the franchisees. In addition, there is

recurring revenue in terms of royalty and platform charges. The franchisees can manage these machines without visiting the physical location. The machines are fully IoT-enabled and connected to the cloud, allowing franchisees to track sales and manage inventory remotely. This machine can produce between 700 and 900 glasses per day. Additionally, they generate income by selling bagasse, the sugarcane waste left after juice extraction.

Having recently launched their machines commercially in Pune, the company aims to achieve revenue of 400 million by 2027, concludes Milind.

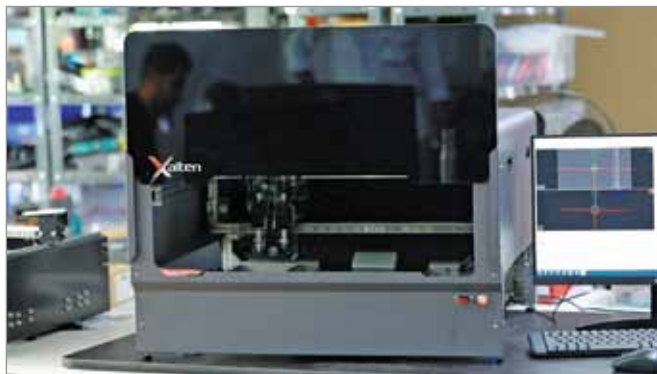
—Nitisha Dubey

2 PRECISION PCB ASSEMBLY WITH XALTAN SYSTEMS’ MADE-IN-INDIA TECH

As of 2024, this startup is recognised for its innovative desktop pick-and-place machine, transforming electronic prototyping and small-scale manufacturing.

Efficiency, accuracy, and adaptability are the future of prototyping and small-scale manufacturing. Dr Unnikrishnan emphasises their desktop pick-and-place machine is a development for engineers, enabling swift idea-to-product transformation. Its capability to manage loose components and customisation makes it perfect for speeding up R&D and small-batch production, empowering users with greater control.

Headquartered in Kochi, India, Xaltan Systems was incorporated in 2023 by three co-founders, Dr Unnikrishnan K.C., Ajith Narayanan, and Suresh Gopinathan. (The groundwork had been ongoing since 2017.) Their flagship product, described as a ‘robotic arm for



XP300

electronics,’ significantly streamlines the process of placing tiny components on circuit boards, offering a precise and adaptable solution, as Unnikrishnan states.

The startup’s pick-and-place machine stands out due to its advanced vision systems and accurate motion systems, allowing it to handle components as small as 0.3mm by

0.5mm with pinpoint accuracy. Its customisable feeders are designed to handle an extraordinary range of components, from tiny resistors and capacitors to larger IC packages, ensuring seamless transitions between different board designs. A key innovation is its Loose Part Feeder system, a rare feature in similar machines, particularly

those from Chinese competitors. This machine was developed amidst the global chip shortage and the challenges posed by the Covid-19 pandemic, which forced the team to overcome significant design hurdles, including eliminating costly external assembly reliance.

The startup manufactures its products in-house at its Kochi